Katta Sai Teja Reddy

es23btech11021@iith.ac.in | saitejareddykatta@gmail.com | +91 9346273293 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.45
NSR IMPULSE JR COLLEGE, HYDERABAD	Class XII (Telangana State Board of Intermediate Education)	2023	98.50%
KESHAVAREDDY CONCEPT SCHOOL , SANGA REDDY	Class X (Board of Secondary Education Telangana)	2021	100.00%

SKILLS

- **Programming Languages:** C, C++, Python
- **Web Technologies:** HTML, CSS, JavaScript, React, Node.js, Express.js
- **Databases:** MongoDB, MySQL
- **Libraries/Tools:** NumPy, Matplotlib, SciPy, Jupyter Notebook, Streamlit , Scikit-learn

RELEVANT COURSES

Computer Science: Data Structures and Applications, Algorithms, Computer Networks, Software Engineering, Database Management Systems, Operating Systems, Theory of Computation, Discrete Mathematics.

Artificial Intelligence: Machine Learning, Fraud Analytics, AI for Finance

Mathematics: Convex Optimisation, Probability and Random Variables, Linear Algebra, Statistics, Calculus, Differential Equations, Transform Techniques, Complex Variables

PROJECTS

TCP Socket-Based Quiz System with Network Protocol Design

C++, TCP Sockets, Multithreading, Client-Server Networking

- Built a **real-time client-server quiz application** in C++ using **TCP sockets** and a custom text-based protocol for communication, session handling, and leaderboard interactions
- Designed and implemented networked communication flows for **multi-client interaction**, including connection handling, request parsing, and response management
- Integrated LLM-generated quiz content into the server flow while emphasizing scalable **socket programming, concurrency**, and protocol-driven message exchange

From-Scratch Machine Learning & Neural Network Library - (AI2000)

Machine Learning, Mathematical Modeling, Neural Networks, Model Selection

- Implemented core **ML algorithms from first principles**, including MLE/MAP estimation, polynomial regression, PCA, K-Means, and GMM with EM, deriving and validating their mathematical formulations.
- Built a reusable **mini-scikit-learn library and autograd engine** with Logistic Regression, preprocessing, K-Fold validation, automatic differentiation, neural-network layers, SGD, and Cross-Entropy loss.
- Performed systematic **model selection and hyperparameter experimentation** on Spambase and Fashion-MNIST,

comparing linear, feature-engineered, and MLP models using validation-based tuning and held-out evaluation.
- Analyzed **generalization, overfitting, feature scaling, dimensionality reduction, and model convergence** through quantitative experiments and comparative evaluation.

AI-Powered Adaptive Authentication & IAM Gateway

Tech: Node.js, Express, MongoDB Atlas, React, JWT, Passport.js

- Built a centralized **SSO/MFA gateway** serving **5,000+ students** across institute-wide applications, supporting per-app configurable auth methods (**local, OAuth2, SAML2, OIDC**) with **MongoDB TTL-backed session management**.
- Designed a **risk-scoring model** using login behavior, geolocation, and **VPN/proxy detection** to dynamically bypass MFA for low-anomaly sessions and enforce step-up verification for high-anomaly ones.
- Engineered auth security infrastructure with **JWT access/refresh tokens**, **bcrypt hashed credential storage**, scheduled key rotation with backward validation, and **CIDR-based IP whitelisting**.
- Implemented **immutable audit logging** across **13+ security event types** to support traceability and post-incident review.

Dataset Distillation using LoRA and PEFT Methods

Machine Learning, Deep Learning, Dataset Distillation, Parameter-Efficient Fine-Tuning

- Conducted extensive research on **dataset distillation and PEFT** by integrating LoRA with dataset distillation paradigms on MobileNetV2 architectures.
- Implemented and evaluated dataset distillation frameworks including **meta-learning, gradient matching, trajectory matching (MTT), and SRe2L** statistical recovery on CIFAR-10 using custom LoRA-based convolutional layers.
- Developed a comprehensive experimental pipeline involving **LoRA integration, gradient-tracking ablations, synthetic dataset optimization, and cross-evaluation** analysis to study the compatibility of low-rank adaptation with dataset distillation techniques.
- Analyzed performance degradation and representational instability in **low-rank constrained models**, demonstrating limitations of combining LoRA with dataset distillation.

POSITIONS OF RESPONSIBILITY

- **Events Coordinator of MILAN:** The General Championship of IIT Hyderabad.

Responsible for planning, coordinating, and executing key Scitech events as part of fest.

- **Logistics Coordinator** in **Media Council**